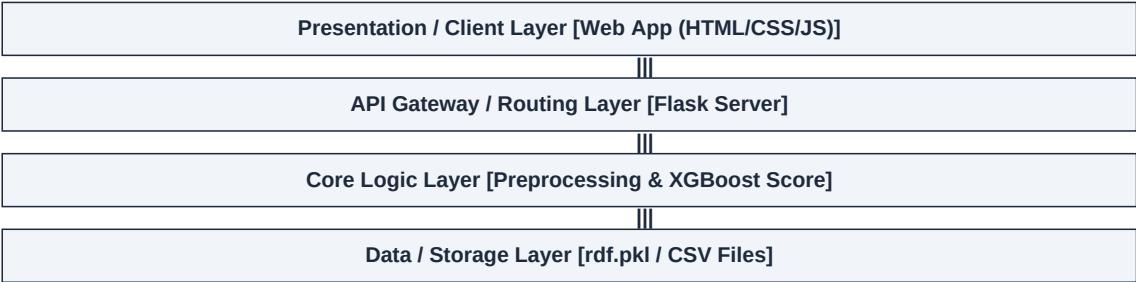


Solution Architecture Specification

Date	06 July 2026	Project Name	Smart Lender
Team Lead	Alla Umesh	Team Members	DSP, JRJ, LPB, SVB

Solution Architecture Layers



Layer Component	Description / Role in Architecture	Technologies Used
Presentation Layer	Responsive dashboard layout rendering forms and CSV drop zones	HTML5, CSS3, Vanilla JS, Google Fonts
API Gateway Layer	Exposes API routes, parses JSON payloads and CSV multipart streams	Flask HTTP server framework
Core Logic Layer	Imputes missing values, encodes categories, log transforms, scales, and runs XGBoost Classifier	Python, scikit-learn, XGBoost, Pandas, Numpy
Data Storage Layer	Manages model pickles, dataset records, and temp CSV outputs	CSV, Pickle files (rdf.pkl)